

7a	Echo	Reverberation
	Reflection from distant surface	Reflection from near surfaces
	Single reflection	Multiple reflections
	Distinct sound	Garbled/overlapping/mixed sound
	Any one of the above.	
7b	With reverberation, speech becomes muddled / can't hear clearly because the reverberations overlapped/interfere with the sound waves generated either by speaker or the reflected waves.	
7ci	Energy is absorbed by the panels.	
7cii	When frequency of sound is equal to natural frequencies of the panel, resonance occurs. Maximum energy is transferred from the sound to the panel, and hence less energy is reflected off the walls, thus reducing reverberation times.	
7d	Other factors that would affect reverberation time: 1. Size/volume of room 2. surface area of absorbing area 3. distance between source and reflecting surface 4. texture of reflecting surface (soft/hard, smooth/rough) 5. room that is filled with objects that can absorb sound 6. Intensity of sound 7. temperature/pressure/density/humidity of air	

7ei	$f = 1280 \text{ Hz}, \frac{1}{f^2} = 0.61 \times 10^{-6} \text{ s}^2$
7eii	1. Point plotted correctly 2. Balanced line drawn
7eiii 1	$\text{Gradient} = \frac{(0.700 - 0.400) \times 10^{-6}}{(5.00 - 1.30)}$ $= 0.0811 \times 10^{-6} \text{ s}^2 \text{ mm}^{-1}$
7eiii 2	subst $(5.00, 0.700 \times 10^{-6})$ in $y = mx + c$, $0.700 \times 10^{-6} = 0.0811 \times 10^{-6} (5.00) + c$ $c = 2.95 \times 10^{-7} \text{ s}^2$
7eiv 1	$f = 5000 \sqrt{\frac{P}{L(t + 0.8d)}}$ $\frac{1}{f^2} = \left(\frac{0.8L}{5000^2 P} \right) d + \frac{Lt}{5000^2 P}$ $\text{Gradient} = \frac{0.8L}{5000^2 P}, \text{ sub } P = 10$ $L = 25.3 \text{ mm}$
7eiv 2	$\frac{Lt}{5000^2 P} = 2.95 \times 10^{-7}$ $t = 2.92 \text{ mm}$
7fi	From Fig. 7.3, resonant frequency for 5.0 mm hole is 1180 Hz. From Fig. 7.5, it can be seen that resonant frequencies have been <u>reduced</u> to 1000 Hz.
7fii	Absorber 2 will be more suitable since it has a higher coefficient of absorption than absorber 1 at all frequencies, implying more energy is absorbed for every frequency and therefore less reverberations.